A MACHINE-CHECKED REDUCTION OF THE LONELY RUNNER CONJECTURE TO A SINGLE CRUX, AND FIRST-ORDER CERTIFICATES

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ABSTRACT. For positive integer speeds $v_1, \dots, v_m$ let $\mathrm{ML}(v) = \sup_t \min_i \|v_i t\|$. The lonely runner conjecture asserts $\mathrm{ML}(v) \geq 1/(m+1)$. We give a proof, checked in Lean 4 with Mathlib, that the conjecture for all m follows from three finite-range statements, each a consequence of published or announced cases of the conjecture, together with one statement FloorCruxU: the bound for primitive families with at least 12 speeds that have primitive leave-one-out subfamilies, contain a multiple of every $j \in [2, m+1]$, carry a $\{-1,0,1\}$-relation, and admit no first-moment certificate with a Riesz-product weight. We also verify that, given the three finite statements, FloorCruxU is equivalent to the conjecture; the reduction localises the problem and does not weaken it. Further verified results: a first-moment criterion for integer families with relations, in the form introduced by Bedert; the fact that no absolutely continuous weight certifies a tight family; a one-sided perturbation lemma at the times $a/(m+1)$; and the bound $\mathrm{ML} \geq (m+1)/(mj+m+1)$ for the families $([1,m] \setminus \{j\}) \cup \{j(m+1)\}$, $(m+1)/2 < j \leq m$, $\gcd(j, m+1) = 1$. For the one-entry families $([1,m]\setminus\{r\})\cup\{w\}$ we verify two sufficient conditions for $\mathrm{ML} \geq 2/(2m+1)$, one at the time $a/(2m+1)$ with $ar \equiv 1$ and one at the time $1/r$; the second is the argument of Goddyn and Wong. We state conjectures on primitive families with a speed divisible by $m+1$, with the exact computational ground, and record examples showing that the denominator of a certifying time is not bounded by any affine function of m tested.

1. INTRODUCTION

1.1. **Conventions.** Throughout, $\|x\|$ is the distance from $x \in \mathbb{R}$ to the nearest integer. A *family* is a tuple $v = (v_1, \dots, v_m)$ of integers; m is the number of *speeds*, and the corresponding runner problem has $m+1$ runners. We put

$$\mathrm{gap}(v,t) = \min_{1 \leq i \leq m} \|v_i t\|, \qquad \mathrm{ML}(v) = \sup_{t \in \mathbb{R}} \mathrm{gap}(v,t).$$

The *target* for m speeds is $1/(m+1)$. For $m \geq 1$ let $\mathrm{LRC}(m)$ denote the statement

$$\mathrm{ML}(v) \geq \frac{1}{m+1} \quad \text{for every } v \in \mathbb{Z}_{>0}^m.$$

Repeated speeds are allowed. This is equivalent to the usual formulation with distinct speeds, since a family with $m' < m$ distinct values has the larger target $1/(m'+1)$. A family is *tight* if $\mathrm{ML}(v) = 1/(m+1)$.

In the Lean development a family with m speeds is a function $\texttt{Fin}\,(k+1) \to \mathbb{Z}$ with $k = m-1$, and the target is written $1/(k+2)$. We translate every Lean statement to the index m; the correspondence is recorded in the file `LEAN_MAP.md` accompanying this paper, together with the output of `#print axioms` for every declaration cited. Every result labelled *Lean* below builds with no `sorry` and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Statements without a Lean name are either cited from the literature or explicitly labelled as conjectures, computations, or unformalized remarks.

1.2. **Results.** Section 3 proves (Theorem 3.3) that LRC(m) for all $m \geq 3$ follows from three finite-range atoms A1, A2, A3 and one crux FloorCruxU, and (Theorem 3.4) that under A1–A3 the crux is equivalent to LRC(m) for all $m \geq 3$. The atoms concern at most 11 speeds; they are consequences of published or announced cases of the conjecture [7, 6, 2, 15, 16, 18, 17] but are not formalized here. The conjecture is known, in the literature, for at most 13 runners [17] (at most 14 by the announcement [1]); the crux restricted to $m \leq 12$ speeds is therefore known, and its open part is exactly the open part of the conjecture.

Section 4 treats first-moment certificates. The soundness of an arbitrary measure as a certificate is Bedert's Lemma 2.1 [4]. We verify a criterion for families with $\{-1, 0, 1\}$-relations (Theorem 4.3) which uses Bedert's Riesz product with an exact normaliser, and a barrier (Theorem 4.6): if $\mathrm{ML}(v) \leq \delta$ the set of times with gap at least δ is countable, so no weight absolutely continuous with respect to Lebesgue measure certifies a tight family. The barrier is elementary and we regard it as folklore. A related device appears in the proof of [3, Thm. 2.1], where the absence of a time with gap strictly above the target makes a product of window functions vanish identically. That a family with no strict lonely time satisfies a bounded integer relation is the relation theorem of [3], building on Czerwinski [8] and Giri and Kravitz [10]; no relation theorem is claimed here.

Section 5 treats certificates at a rational base point a/q shifted by a common small time ε. We verify a one-sided perturbation lemma (Theorem 5.2) and the bound $\mathrm{ML}(F(m, j)) \geq (m + 1)/(mj + m + 1)$ for the families $F(m, j) = ([1, m] \setminus \{j\}) \cup \{j(m + 1)\}$ (Theorem 5.3). For $j = m$ the value is Kravitz's exact value [12, Thm. 5.1]; for $j < m$ the strict inequality $\mathrm{ML}(F(m, j)) > 1/(m + 1)$ follows from Zhang's classification [19, Thm. 1.2] combined with the Goddyn–Wong condition [11]. What is added is the explicit bound, its witness, and the formalization.

Section 6 states Conjectures 6.1–6.4 with their computational ground. Section 7 proves (T2) for one-entry modifications of $[1, m]$ under explicit residue conditions (Theorems 7.1 and 7.3), which cover all but 42 of the 15 307 primitive such families with $3 \leq m \leq 40$, $w \leq 20m$ and $(m+1) \mid w$; the remaining 42 satisfy (T2) by exact computation. Section 8 records, briefly, why the certificates of Sections 4 and 5 do not by themselves reach the crux.

2. Classes of families

Definition 2.1. Let $v \in \mathbb{Z}^m_{>0}$.

(1) v is *primitive* if $\gcd(v_1, \ldots, v_m) = 1$.
(2) v is *leave-one-out primitive* if for every l the family $(v_i)_{i \neq l}$ is primitive.
(3) v is *divisor-complete* if every integer j with $2 \leq j \leq m + 1$ divides some v_i.
(4) The *web* of v is $W(v) = \{\sigma \in \{-1, 0, 1\}^m \setminus \{0\} : \sum_i \sigma_i v_i = 0\}$. The family is *strongly free* if $W(v) = \emptyset$ (Lean: `StrongFree`).
(5) v is *two-free* if for every i the family obtained by replacing v_i by $2v_i$ is strongly free (Lean: `TwoFree`).

Definition 2.2 (Riesz-product union bounds)**.** For $v \in \mathbb{Z}^m$ and $\delta \in \mathbb{R}$ let $B_i(\delta) = \{t \in \mathbb{R} : \|v_i t\| < \delta\}$ (Lean: `badSet`) and $P_v(t) = \prod_{i=1}^m (1 - \cos 2\pi v_i t)$. For $D \in \mathbb{N}$ we say that U_D *fires on* v (Lean: `UDFires`) if

$$\sum_{i=1}^{m} \int_{[0,1] \cap B_i(1/(m+1))} P_v(t)^D \, dt \;<\; \int_{[0,1]} P_v(t)^D \, dt.$$

For $D = 0$ this is the Lebesgue union bound.

3. The reduction

3.1. **The atoms and the crux.** The following four statements are Lean `Prop`s; none is proved in the development.

Definition 3.1 (Atoms)**.** A1 (Lean: `StrongFreeWindowTiny`.) For $4 \le m \le 7$, every strongly free $v \in \mathbb{Z}^m_{>0}$ satisfies $\mathrm{ML}(v) \ge 1/(m+1)$.

A2 (Lean: `DissociatedOctetWitness`.) For every strongly free $v \in \mathbb{Z}^8_{>0}$ that is not two-free there are integers $p \ge 0$, $q \ge 1$ such that for all i, with r_i the least nonnegative residue of $v_i p$ modulo q,

$$q \;\le\; 9\min(r_i,\; q - r_i),$$

that is, $\|v_i p/q\| \ge 1/9$.

A3 (Lean: hypothesis `hwinB`.) For $1 \le m \le 11$, every $v \in \mathbb{Z}^m_{>0}$ that is not strongly free satisfies $\mathrm{ML}(v) \ge 1/(m+1)$.

Definition 3.2 (The crux; Lean: `RoomA4Compress.FloorCruxU`)**.** FloorCruxU is the statement: for every $m \ge 12$, if $\mathrm{LRC}(m')$ holds for all $1 \le m' < m$, then every $v \in \mathbb{Z}^m_{>0}$ that is primitive, leave-one-out primitive, divisor-complete and not strongly free, and on which U_D fires for no $D \in \mathbb{N}$, satisfies $\mathrm{ML}(v) \ge 1/(m+1)$.

Theorem 3.3 (Lean: `RoomA4Compress.universalFloor_of_four_atoms_U`)**.** A1 $\wedge$ A2 $\wedge$ A3 $\wedge$ FloorCruxU *implies* $\mathrm{LRC}(m)$ *for every* $m \ge 3$.

The proof is a strong induction on m. Strongly free families are handled by a ladder argument (Lean: `separationSF_of_tiny_and_residual`, `ladderSF_le`) which uses A1 and A2; families that are not strongly free and have $m \le 11$ are A3. For $m \ge 12$ the induction step applies, in order, Lemma 3.8, Lemma 3.7, Proposition 4.2, and finally the crux. Scale invariance reduces to primitive families.

Theorem 3.4 (Lean: `RoomFCUErdos.floorCruxU_iff_universalFloor`)**.** *Assume* A1*,* A2 *and* A3*. Then* FloorCruxU *holds if and only if* $\mathrm{LRC}(m)$ *holds for every* $m \ge 3$.

The reverse implication is immediate, since the conclusion of the crux is an instance of $\mathrm{LRC}(m)$. Theorem 3.4 says that the reduction loses nothing and gains nothing: the crux carries the whole difficulty of the conjecture at $m \ge 12$ speeds.

Proposition 3.5 (Lean: `strongFreeWindowTiny_of_456`)**.** *A1 follows from its restriction to* $5 \le m \le 7$*. The case* $m = 4$ *is the verified five-runner theorem.*

Remark 3.6 (Status of the atoms). None of A1 ($m = 5, 6, 7$), A2, A3 is formalized. Each follows from known cases of the conjecture. A1 and A3 are instances of $\mathrm{LRC}(m)$ for $m \le 7$ and $m \le 11$ respectively; these are due to Cusick and Pomerance [7] ($m = 4$), Bohman, Holzman and Kleitman [6] ($m = 5$), Barajas and Serra [2] ($m = 6$), Rosenfeld [15] ($m = 7$), Rosenfeld [16] and Trakulthongchai [18] ($m = 8, 9$), and Sungkawichai and Trakulthongchai [17] ($m = 10, 11$; preprint). A2 follows from LRC(8) together with the fact that for rational δ the set $\{t : \mathrm{gap}(v, t) \ge \delta\}$ is a finite union of closed intervals per period with rational endpoints, so that $\mathrm{ML}(v) \ge \delta$ is witnessed by a rational time; this last step is standard and is not formalized. The case $\mathrm{ML}(v) > \delta$ is Lean `RoomFCUCompress.rational_point_of_lt`. Consequently Theorem 3.3 is a reduction of the conjecture to the crux modulo the cited literature, and a Lean theorem only with A1–A3 as hypotheses.

3.2. **Uniform lemmas used in the step.** The following two lemmas are taken from Rosenfeld [15]. Lemma 3.8 is his Lemma 5. His Lemma 4 states that a family without the lonely runner property has $\mathrm{lcm}(2, \ldots, m+1) \mid \prod_i v_i$; Lemma 3.7 is the time-$1/j$ argument behind it, in the form in which j divides a single speed.

Lemma 3.7 (Lean: `RoomA4Rosenfeld.witness_of_missing_divisor`, `RoomFCUErdos.floor_of_not_D`)**.** *Let $v \in \mathbb{Z}^m_{>0}$ and $2 \le j \le m+1$ with $j \nmid v_i$ for all i. Then $q = j$, $p = 1$ satisfy $q \le (m+1)\min(r_i, q - r_i)$ for all i, where $r_i \equiv v_i \pmod j$, $0 \le r_i < j$. In particular, if $m+1$ divides no v_i then $\mathrm{ML}(v) \ge 1/(m+1)$.*

Lemma 3.8 (Lean: `RoomA4Rosenfeld.floor_of_leaveOneOut_imprimitive`)**.** *Let $m \ge 3$, assume $\mathrm{LRC}(m-1)$, and let $v \in \mathbb{Z}^m_{>0}$ be primitive with $\gcd\big((v_i)_{i\ne l}\big) \ne 1$ for some l. Then $\mathrm{ML}(v) \ge 1/(m+1)$.*

Let $\mathcal{D}$ be the set of families $v \in \mathbb{Z}^m_{>0}$, $m \ge 3$, in which $m+1$ divides some speed.

Proposition 3.9 (Lean: `RoomFCUErdos.universalFloor_iff_divisible`)**.** $\mathrm{LRC}(m)$ *for all $m \ge 3$ is equivalent to the bound $\mathrm{ML}(v) \ge 1/(m+1)$ for all $v \in \mathcal{D}$.*

Proposition 3.10 (Lean: `RoomFCUModerator.strictFloorD_false`)**.** *The strict bound $\mathrm{ML}(v) > 1/(m+1)$ fails on $\mathcal{D}$: $v = (4, 8, 12)$ has $\mathrm{ML}(v) = 1/4$.*

Proposition 3.10 is why Conjectures 6.1 and 6.4 below carry the hypothesis gcd = 1.

Remark 3.11. The crux is the weakest of several forms of the last atom recorded in the development. The earlier forms `BridgeDisjunctionQPrim` and `RoomA4Rosenfeld.BridgeDisjunctionQPrimInd2` ask, on successively smaller classes, for either the criterion of Theorem 4.3 or a rational witness; each implies FloorCruxU (Lean: `RoomA4Rosenfeld.bridgeDisjunctionQPrimInd_of_QPrim`, `RoomA4Rosenfeld.ind_of_ind2`, `RoomA4Compress.floorCruxU_of_ind2`). Theorem 3.4 shows that no further weakening of this type can make the crux easier than the conjecture.

4. First-moment certificates

Proposition 4.1 (Lean: `RoomA4Compress.floor_of_measure_certificate`)**.** *Let $v \in \mathbb{Z}^m$, $\delta \in \mathbb{R}$ and let ν be any measure on $\mathbb{R}$. If $\sum_i \nu(B_i(\delta)) < \nu(\mathbb{R})$, then $\delta \le \mathrm{ML}(v)$.*

This is Bedert's Lemma 2.1 [4], stated for an arbitrary measure. A Dirac mass recovers rational witnesses, counting measure on a coset $t_0 + q^{-1}\mathbb{Z}$ recovers Lemma 3.8-type arguments, and a weight recovers the following.

Proposition 4.2 (Lean: `RoomA4Compress.floor_of_integral_unionBound`, `RoomA4Compress.floor_of_UD`)**.** *Let $v \in \mathbb{Z}^m$, $\delta \in \mathbb{R}$, and $F : \mathbb{R} \to \mathbb{R}$ continuous and nonnegative. If $\sum_i \int_{[0,1]\cap B_i(\delta)} F < \int_{[0,1]} F$ then $\delta \le \mathrm{ML}(v)$. In particular, if U_D fires on v for some $D \in \mathbb{N}$ then $\mathrm{ML}(v) \ge 1/(m+1)$.*

4.1. **The web criterion.** For $v \in \mathbb{Z}^m$ define the rationals

$$M(v) = 1 + \sum_{\sigma \in W(v)} \Big(-\frac{1}{2}\Big)^{|\operatorname{supp}\sigma|}, \qquad M_l(v) = 1 + \sum_{\substack{\sigma \in W(v) \\ \sigma_l = 0}} \Big(-\frac{1}{2}\Big)^{|\operatorname{supp}\sigma|}$$

(Lean: `webMassQ`, `webMassDelQ`). $M(v)$ is the constant Fourier coefficient of P_v, and $M_l(v)$ that of $P_{(v_i)_{i\ne l}}$.

Theorem 4.3 (Lean: `floor_of_rieszSharpFiresQ`)**.** *Let $v \in \mathbb{Z}^m$. If $M(v) > 0$ and*

$$99225\Big(\sum_{l=1}^{m} \big(2M_l(v) - M(v)\big) + M(v)\Big) \;\le\; 10000\,(m+1)^2\,M(v),$$

then $\mathrm{ML}(v) \ge 1/(m+1)$.

The hypothesis is the predicate `rieszSharpFiresQ`; the constant $99225/10000$ exceeds π^2. The proof applies Proposition 4.1 to the probability measure with density $P_v/M(v)$ and bounds the mass of each B_l through the subfamily coefficient $M_l(v)$. The method, including the expansion of P_v over $\{-1,0,1\}^m$ with weights $(-1/2)^{|\operatorname{supp}\sigma|}$, is Bedert's [4, Lemma 4.1], where v is dissociated and $M(v)=1$. The additions are the normaliser $M(v)$ for families with relations and the exact per-coordinate term in place of a supremum bound.

Corollary 4.4 (Lean: `rieszSharpFiresQ_of_subfamilySum_le`)**.** *Let $m \geq 9$ and $v \in \mathbb{Z}^m$ with $M(v) > 0$ and $\sum_l M_l(v) \leq m\,M(v)$. Then the hypothesis of Theorem 4.3 holds, and $\mathrm{ML}(v) \geq 1/(m+1)$.*

Remark 4.5. If v is strongly free then $W(v)=\emptyset$, so $M(v)=M_l(v)=1$ and Corollary 4.4 gives $\mathrm{ML}(v)\geq 1/(m+1)$ for strongly free families with $m\geq 9$. The identification $W(v)=\emptyset$ is immediate from the definitions and is not a separate Lean declaration. Bedert's supremum bound, read at the target $1/(m+1)$, gives the same conclusion from $m=18$.

4.2. **The barrier.** For $\delta\in\mathbb{R}$ put $G_\delta(v)=\{t\in\mathbb{R}:\ \|v_it\|\geq\delta \text{ for all } i\}$.

Theorem 4.6. *Let $v\in\mathbb{Z}^m$ with all $v_i\neq 0$.*

1. *(Lean: `RoomA4Barrier.goodSet_countable`.) If $\mathrm{ML}(v)\leq\delta$ then $G_\delta(v)$ is countable.*
2. *(Lean: `RoomA4Barrier.integral_le_sum_badSet`.) If $\mathrm{ML}(v)\leq\delta$ and $F\geq 0$ is integrable on $[0,1]$, then $\int_{[0,1]}F\leq\sum_i\int_{[0,1]\cap B_i(\delta)}F$.*
3. *(Lean: `RoomA4Barrier.lt_ML_of_UDFires`.) If U_D fires on v then $\mathrm{ML}(v)>1/(m+1)$.*
4. *(Lean: `RoomA4Barrier.not_UDFires_of_tight`.) If v is tight, U_D fires on v for no D.*
5. *(Lean: `RoomA4Barrier.not_UDFires_consec`.) U_D fires on $(1,2,\dots,m)$ for no D.*

Thus every tight family satisfies the last hypothesis of the crux, and Proposition 4.2 only ever certifies the strict inequality. Whether a tight family lies in the class of Definition 3.2 is open; Conjecture 6.1 would exclude it.

5. Escape certificates

Proposition 5.1 (Lean: `RoomFCUModerator.ML_ge_of_escape`)**.** *Let $v\in\mathbb{Z}^m$, $q,a\in\mathbb{Z}$ with $q>0$, and $\varepsilon,\delta\in\mathbb{R}$ with $\delta>0$. Suppose that for every i there are $N_i,r_i\in\mathbb{Z}$ with $v_ia=qN_i+r_i$ and*

$$\delta\ \leq\ \frac{r_i}{q}+v_i\varepsilon\ \leq\ 1-\delta.$$

Then $\delta\leq\mathrm{ML}(v)$.

The time used is $a/q+\varepsilon$. The statement is elementary and the pattern is standard [12]. With $\varepsilon=0$ and $\delta=A/B$ it takes a purely arithmetic form (Lean: `RoomFCUCompress.ML_ge_of_residues`). Its converse also holds with $q=1$ and ε a maximiser (Lean: `RoomFCUCompress.lonely_iff_escape`); the proposition therefore has content only together with a bound on q, which is the subject of Section 8.

Theorem 5.2 (Lemma P; Lean: `RoomFCUCompress.lemmaP_quant`, `RoomFCULemmaP.lemmaP`)**.** *Let $m\geq 3$, $q=m+1$ and $v\in\mathbb{Z}^m_{>0}$. Suppose there is an index i_0 such that*

1. *$q\mid v_{i_0}$;*
2. *for every $i\neq i_0$, $q\nmid v_i$ and $v_i<v_{i_0}$;*
3. *there is u with $\gcd(u,q)=1$ and $v_i\not\equiv u \pmod q$ for all i.*

Put $W=v_{i_0}$. Then

$$\mathrm{ML}(v)\ \geq\ \frac{2W}{q(2W-1)}\ >\ \frac{1}{m+1}.$$

The witness is $t = a/q - 2/(q(2W-1))$ with $au \equiv 1 \pmod q$: the largest speed moves from 0 to distance $2W/(q(2W-1))$, while every other speed has residue $r \in [2, q-2]$ at a/q and moves by less than the available margin. We found no statement of this lemma in the literature consulted ([12, 19, 13]).

Theorem 5.3 (Lean: `RoomFCUCompress.famF_via_escape`, `RoomFCURamanujan.famF_lonely`)**.** *Let m, j be positive integers with $(m+1)/2 < j \le m$ and $\gcd(j, m+1) = 1$, and put $F(m,j) = ([1,m] \setminus \{j\}) \cup \{j(m+1)\}$ and $D = mj + m + 1$. There is an integer $s \ge 0$ such that the time $t = (sD+1)/(jD)$ satisfies $\|xt\| \ge (m+1)/D$ for every $x \in F(m,j)$. Consequently every family v all of whose speeds lie in $F(m,j)$ satisfies*

$$\mathrm{ML}(v) \;\ge\; \frac{m+1}{mj+m+1} \;>\; \frac{1}{m+1}.$$

For $j = m$, $F(m,m) = \{1, \dots, m-1, m(m+1)\}$ and the bound is Kravitz's value $\mathrm{ML}(1, \dots, n-1, ns) = s/(ns+1)$ at $n = m$, $s = m+1$, where equality holds [12, Thm. 5.1]. For $j < m$, $F(m,j)$ is $[n-1]$ with $n = m+1$ and the speed $r = j$ replaced by μr, $\mu = m+1$. Goddyn and Wong [11, Thm. 2.3] show that such a family is tight only if $\gcd(r,b) > 1$ for every b with $n-r \le b < \mu(n-r)$; here $b = n-r = m+1-j$ gives $\gcd(j, m+1-j) = \gcd(j, m+1) = 1$, so $\mathrm{ML}(F(m,j)) > 1/(m+1)$ was known (see also Zhang's classification [19, Thm. 1.2]). Theorem 5.3 makes this quantitative. The value has the shape $s/(ns+k)$ with $(n,s,k) = (j, m+1, m+1-j)$, the shape of the amended spectrum conjecture of Fan and Sun [9]; we have not checked that their parameter ranges cover these values.

Conjecture 5.4. *Under the hypotheses of Theorem 5.3,* $\mathrm{ML}(F(m,j)) = (m+1)/(mj+m+1)$.

Ground: exact rational computation of $\mathrm{ML}(F(m,j))$ agrees in all 499 admissible pairs with $m \le 56$ (Appendix A).

Proposition 5.5 (Lean: `RoomFCUGoddyn.fast_runner_floor`, `RoomFCUGoddyn.cruxU_fast`)**.** *Let $m \ge 2$, $v \in \mathbb{Z}^m_{>0}$ and l an index such that $mv_i \le v_l$ for all $i \ne l$. If $\mathrm{ML}\left((v_i)_{i \ne l}\right) \ge 1/m$ then $\mathrm{ML}(v) \ge 1/(m+1)$.*

This is the case $L = 1/m$, $\varepsilon = 1/(m(m+1))$ of Kravitz's Proposition 8.1 [12]. It gives the crux for every family with one speed at least m times all others, for example $F(m,m)$, and it is not an instance of Proposition 5.1 with bounded q.

6. Conjectures on the divisible class

All computations in this section use exact integer arithmetic and the standard fact that $\mathrm{ML}(v)$ is attained at a time $c/(v_i + v_j)$ with $i \le j$ and $c \in \mathbb{Z}$; this fact is not formalized. A *box* (m, H) is the set of primitive m-element subsets of $[1, H]$, enumerated completely.

Conjecture 6.1 ((T))**.** *If $v \in \mathbb{Z}^m_{>0}$ is primitive and tight, then $m+1$ divides no v_i.*

Conjecture 6.2 ((T′))**.** *If $v \in \mathbb{Z}^m_{>0}$ is primitive and tight, then every t with $\mathrm{gap}(v,t) = 1/(m+1)$ lies in $\frac{1}{m+1}\mathbb{Z}$.*

(T′) implies (T): at $t = a/(m+1)$ a speed divisible by $m+1$ is at distance 0. For $v = (1, \dots, m)$, (T′) is contained in Kravitz's Proposition 10.4 [12]: the equality times are the $a/(m+1)$ with $\gcd(a, m+1) = 1$. By Lemma 3.7, (T) is equivalent to the statement that no primitive tight family is divisor-complete with $m+1$ in the divisor set; in particular (T) would exclude tight families from the class of the crux. The gcd hypothesis cannot be dropped: $5 \cdot (1,2,3,4)$ is tight and every speed is divisible by 5.

Proposition 6.3 (Lean: `RoomFCUHilbert.gw_not_dvd`, `RoomFCUHilbert.gw_family_not_dvd`)**.** *Let $0 < r < n$ and $\mu \geq 1$ be integers such that $\gcd(r, b) > 1$ for every integer b with $n - r \leq b < \mu(n - r)$. Then $n \nmid r\mu$.*

The hypothesis is the Goddyn–Wong condition for accelerating the runner r of $[n-1]$ to $r\mu$ [11, Thm. 2.3], and the argument is theirs: it is the last step of the proof of [11, Thm. 3.1], which shows $n \nmid m_r r$ for every accelerated speed $m_r r$ in order to use the time $1/n$. We have checked every tight instance listed or constructed in [11] against (T). The sporadic list there (all tight vectors with $n \leq 20$ and maximum speed at most 40: $\{1,3,4,7\}$, $\{1,3,4,5,9\}$, $\{1,4,5,6,7,11,13\}$, $\{1,2,3,4,5,7,12\}$, $[13]_{12\mapsto 24}$, $[19]_{18\mapsto 36}$) has no speed divisible by n; the families of [11, Thms. 2.3, 3.1] have none by the argument just quoted; the exceptional case $(n, r, \mu) = (3, 1, 4)$ of [11, Thm. 2.3] is $(4, 2)$, which is not primitive and has no speed divisible by 3. The two further sporadic tight one-entry families of Zhang [19, Thm. 1.2], $\{1,3,4,7\}$ and $\{1,3,4,5,9\}$, are the first two of this list. Hence no published tight family contradicts (T). The deduction from Proposition 6.3 to those instances rests on the cited theorems and is not formalized.

Conjecture 6.4 (T2)**.** *If $v \in \mathbb{Z}^m_{>0}$, $m \geq 3$, is primitive and $m + 1$ divides some v_i, then* $\mathrm{ML}(v) \geq 2/(2m + 1)$.

Remark 6.5. (T2) implies LRC(m): for $v \in \mathbb{Z}^m_{>0}$ with $g = \gcd(v)$, $\mathrm{ML}(v) = \mathrm{ML}(v/g)$; if $m + 1$ divides some v_i/g apply (T2), otherwise Lemma 3.7. This argument is short and is not formalized. (T2) fails without $\gcd = 1$ (Proposition 3.10) and fails for primitive families outside $\mathcal{D}$: $\mathrm{ML}(1, 5, 6, 11, 16, 17) = 5/33 < 2/13$ (computed).

6.1. **Computational ground.** Table 1 lists the complete enumerations. In every box, every primitive family with $\mathrm{ML}(v) < 1/m$ was retained and its value computed exactly. Since $2/(2m+1) < 1/m$, a counterexample to (T2) in the box would have been retained.

m	H	primitive families	tight families found
3	1500	467 278 456	$[1,3]$
4	500	2 380 658 191	$[1,4]$, $\{1,3,4,7\}$
5	200	2 450 913 356	$[1,5]$, $\{1,3,4,5,9\}$
6	110	2 110 853 351	$[1,6]$
7	60	384 092 920	$[1,7]$, $\{1,2,3,4,5,7,12\}$, $\{1,4,5,6,7,11,13\}$
8	48	376 600 645	$[1,8]$
9	40	273 270 205	$[1,9]$
10	36	254 143 032	$[1,10]$

TABLE 1. Complete enumerations of primitive m-subsets of $[1, H]$.

In all boxes of Table 1: no family has $\mathrm{ML}(v) < 1/(m+1)$; no tight family has a speed divisible by $m + 1$; every maximiser of every tight family has denominator $m + 1$. This is the ground for (T) and (T′). An independent enumeration with a different program (boxes $(4, 400)$, $(5, 200)$, $(6, 100)$, $(7, 64)$, $(8, 50)$, $(9, 42)$, $(10, 38)$) found the same tight families.

For (T2) the minimum of ML over the primitive members of $\mathcal{D}$ in the box is exactly $2/(2m+1)$ for $m = 3, \ldots, 7$, and is $4/33$, $4/37$, $3/31$ for $m = 8, 9, 10$, all above $2/(2m+1)$. An earlier search at $m = 12$ over heights at most 200 reported $\mathrm{ML} = 2/25$ for $(1, 4, 6, 7, 10, 11, 12, 13, 16, 17, 19, 23)$. That value is wrong and the claim that the bound is attained at $m = 12$ is withdrawn: the program that produced it enumerated the times $c/(v_i + v_j)$ with c coprime to $v_i + v_j$ only, and so missed the maximiser $t = 1/9$, whose denominator is not a sum of two speeds. The exact value

is $\mathrm{ML} = 1/9 > 2/25$ (exact enumeration over all numerators, `code/obstructions/mlcheck`; a scan of all $t = p/q$ with $q < 300$ in exact rational arithmetic gives the same value). Equality $\mathrm{ML} = 2/(2m+1)$ in $\mathcal{D}$ is therefore recorded only for $m = 3, \dots, 7$ (Table 1).

For $m = 14, 16, 18, 20$ simulated annealing over primitive members of $\mathcal{D}$ (not complete) ran with the parameters of Table 2. No value below $2/(2m+1)$ was found. At height 5000 the minimum found was $\mathrm{ML}(F(m,m)) = (m+1)/(m^2+m+1)$ in every case.

m	max height	time (s)	evaluations	least $(2m+1)\,\mathrm{ML}$ found
14	5000	720	70 353 611	2.0616 (15/211)
16	5000	720	25 184 829	2.0549 (17/273)
18	5000	720	15 945 569	2.0496 (19/343)
20	5000	720	13 549 085	2.0451 (21/421)
14	60	180	63 550 117	2.0714 (1/14)
16	60	180	48 313 493	2.0625 (1/16)
18	60	180	37 979 054	2.0556 (1/18)
20	60	180	39 240 525	2.0500 (1/20)

TABLE 2. Searches for counterexamples to (T2); one run per row, 4 threads, fixed seeds.

Tight instances in the boxes are few, so the independent evidence for (T) and (T′) is thin. The value $(2m+1)(m+1)/(m^2+m+1)$ tends to 2, so (T2), if true, is asymptotically sharp along $F(m,m)$.

7. (T2) on one-entry families

For $1 \le r \le m$ and $w \in \mathbb{Z}$ let

$$V(m,r,w) = \big([1,m] \setminus \{r\}\big) \cup \{w\},$$

the family $(1, \dots, m)$ with the entry r replaced by w (Lean: `T2Sub.oneEntry`). No hypothesis on w other than those stated is needed below; in particular w may coincide with an entry, and $(m+1) \mid w$ is not assumed.

Theorem 7.1 (Lean: `T2Sub.oneEntry_ML_ge`)**.** *Let $1 \le r \le m$, and let q, a be integers with $q > m$, $q \mid ar-1$, and $q > m+r$ or $q = 2r$. If $w \not\equiv 0,\ r,\ -r \pmod q$, then $\mathrm{ML}(V(m,r,w)) \ge 2/q$.*

Proof. Take $t = a/q$. Since a is inverse to r modulo q, $xa \equiv 0, \pm 1 \pmod q$ holds exactly when $x \equiv 0, \pm r \pmod q$. For $x \in [1,m] \setminus \{r\}$: $0 < x < q$; $x \equiv r$ forces $x = r$ because $|x - r| < q$; and $x \equiv -r$ forces $x + r = q$, impossible if $q > m + r$ and giving $x = r$ if $q = 2r$. For $x = w$ the three congruences are excluded by hypothesis. So every xa has residue in $[2, q-2]$ and $\|xt\| \ge 2/q$. The Lean proof goes through Lean `RoomFCUCompress.ML_ge_of_residues`. □

Corollary 7.2 (Lean: `T2Sub.oneEntry_T2`)**.** *If $1 \le r \le m$, $\gcd(r, 2m+1) = 1$ and $w \not\equiv 0, \pm r \pmod{2m+1}$, then $\mathrm{ML}(V(m,r,w)) \ge 2/(2m+1)$.*

Theorem 7.3 (Lean: `T2Sub.oneEntry_ML_ge_inv`, `T2Sub.oneEntry_T2_inv`)**.** *If $1 \le r \le m < 2r$ and $r \nmid w$, then $\mathrm{ML}(V(m,r,w)) \ge 1/r \ge 2/(2m+1)$.*

Proof. At $t = 1/r$ every speed x has $\|xt\| \ge 1/r$ unless $r \mid x$; the only multiple of r in $[1,m]$ is r, which has been removed, and $r \nmid w$. Finally $2r \le 2m < 2m+1$. □

Theorem 7.3 is not new. It is the first case of the proof of Goddyn and Wong's Lemma 2.2 [11] (a proper subfamily $[n-1] \setminus \{x\}$ with $2x \ge n$ has $\mathrm{ML} \ge 1/x$, by the time $1/x$), and the same time

is used in the proof of Zhang's Lemma 2.3 [19] (if $2r > n-1$ and $[n-1]_{r\to w}$ is tight then $r \mid w$). We include it because it covers residue classes of w that Theorem 7.1 misses. Theorem 7.1 uses the standard base point a/q with q just above m; we did not find it stated in [11, 19, 12], but it is a base-point argument of the type of Lemma 3.7 and Proposition 5.1, and we do not claim it as a new method. Zhang [19, Thm. 2.9] computes $\mathrm{ML}([n-1]\setminus\{2\}) = 2/(2\lfloor n/2\rfloor + 3)$ exactly; Theorem 7.1 gives only a lower bound, for all r.

Combined with the classification of tight one-entry families [19, Thm. 1.2] and Proposition 6.3, no one-entry family with $(m+1) \mid w$ is tight, so (T) holds on this class (a consequence of the cited theorem, not formalized). (T2) asks for the quantitative bound $2/(2m+1)$.

Computational ground. Let $\mathcal{O}$ be the set of triples (m, r, w) with $3 \le m \le 40$, $1 \le r \le m$, $(m+1) \mid w$, $m+1 \le w \le 20m$ and $\gcd V(m, r, w) = 1$; $|\mathcal{O}| = 15\,307$. For every triple we computed $\mathrm{ML}(V(m, r, w))$ exactly (Appendix A). Results:

(1) $\min_{\mathcal{O}}(2m+1)\,\mathrm{ML} = 2$; no member violates (T2).
(2) Equality $\mathrm{ML} = 2/(2m+1)$ holds exactly for $\{1,3,4\}$ ($m=3$), $\{1,3,4,5\}$ ($m=4$) and $\{1,3,4,5,18\}$ ($m=5$), that is, $(m,r,w) = (3,2,4), (4,2,5), (5,2,18)$.
(3) Theorem 7.1 with some $q \in [m+2, 2m+1]$ or Theorem 7.3 applies to all but 42 triples. These are listed in Table 3; for each, (T2) holds by the computed value.

Thus (T2) on one-entry families follows from Theorems 7.1 and 7.3 for every triple satisfying their residue conditions, for all m and without the bound $w \le 20m$; outside those residue classes it is supported by computation only.

8. Obstructions

Remark 8.1 (Base points). Lemma 3.7, Theorem 5.2 and Theorem 5.3 are instances of Proposition 5.1: a base point a/q and a common shift ε, followed by a residue check. By Proposition 5.1 and its converse this form is complete for every family, so all content lies in bounding q in terms of m. Proposition 5.5 is an instance only with $q = 1$, and the first-moment certificates of Section 4 are not of this form; by Theorem 4.6 the latter cannot certify a tight family.

Remark 8.2 (Level drift). Let $m \ge 12$ and let $\mathcal{C}_m$ be the class of Definition 3.2. Computation shows that no affine bound $q \le 2m + c$ with $c \in \{-m+1, 1, 2\}$ suffices, on $\mathcal{C}_m$, for the variants of Proposition 5.1 tested (window certificates at a single level q, first-order shifts, exact scans). Examples (membership in $\mathcal{C}_m$ and all levels computed exactly; not formalized):

- $m = 12$: $(1, 4, 5, 6, 7, 9, 11, 13, 40, 60, 238, 2731)$ has no first-order window certificate with $q \le m + 1 = 13$, even allowing both signs and an exact scan of the multiples of q; the first such level is 22.
- $m = 16$: $(1, 5, 6, 7, 8, 9, 11, 13, 17, 19, 23, 25, 29, 31, 366, 1680)$ has no exact window certificate at a level q with $m + 1 < q \le 2m + 2 = 34$; the first such level is 35. This family has a certificate at $q = 2$, so it bounds only the window variant.
- $m = 18$: $(1, 3, 4, 5, 7, 11, 13, 17, 19, 23, 31, 70, 74, 116, 180, 272, 3986, 4943)$ has no certificate, first-order or exact, at $q \le 2m + 1 = 37$; the first is a point certificate at $q = 38$, and $\mathrm{ML} = 28/191$.
- $m = 18$: $(1, 3, 4, 5, 7, 11, 13, 16, 17, 19, 23, 25, 29, 31, 108, 185, 210, 3685)$ has its first certificate at $q = 46 > 2m + 2$.

In each case a certificate exists at a larger level, so $\mathrm{ML}(v) \ge 1/(m+1)$. The families consist of a small core together with two or three large speeds, and the level at which a certificate first appears was not predicted by any of the invariants tested (Appendix A).

Proposition 8.3 (Lean: `RoomFCUMss.halfShift_not_lonely`). *There is no $t \in \mathbb{R}$ with*

$$\|t\|,\ \left\|2t-\tfrac{1}{2}\right\|,\ \left\|3t-\tfrac{1}{2}\right\|,\ \left\|4t-\tfrac{1}{2}\right\|,\ \left\|6t-\tfrac{1}{2}\right\|,\ \|9t\| \ \geq\ \tfrac{1}{7}.$$

With zero phases the speeds $(1,2,3,4,6,9)$ satisfy LRC(6) [2]. Hence an argument for the conjecture must use that the starting phases are all zero, not merely that they lie in $\frac{1}{2}\mathbb{Z}$. Counterexamples to the conjecture with general phases are due to Blanco, Criado and Santos [5], from $n = 5$ in their indexing; see also [14] for their growth. Only the instance with phases in $\{0, \frac{1}{2}\}$ is not taken from the literature.

Appendix A. Computational ground and reproduction

All programs are in Rust with exact integer arithmetic and are in `code/reduction/<name>/`; the file `code/reduction/README.md` gives the exact command reproducing each number.

- Table 1, maximiser denominators, (T2) minima for $m \leq 10$: `fcuspec`. Independent enumeration: `gdn`; the smaller boxes of a third program `ecalc` for $m \leq 8$ agree.
- Table 2: `t2hunt`, seeds $101m+1$ (height 5000) and $101m+2$ (height 60); logs record the reported minima.
- Conjecture 5.4: exact ML$(F(m,j))$ for all admissible (m,j), $m \leq 56$: `rmj`.
- Level-drift examples: `mdr`, an independent checker; the values of ML used in the text are recomputed by `code/obstructions/mlcheck`, which enumerates all numerators (the function `ml` of `mdr` originally skipped numerators not coprime to the denominator and could return a value below ML; the archived source is corrected); re-checked with `modchk`. Invariant fits of the certificate levels: `levelfit`.
- Section 7: `t2sweep` (sweep of $\mathcal{O}$, coverage by Theorems 7.1 and 7.3, Table 3).

m	r	w	ML	m	r	w	ML	m	r	w	ML
3	1	36	5/13	4	4	40	10/41	7	7	112	16/113
3	2	12	2/5	5	2	90	10/47	8	6	108	18/113
3	2	16	8/19	5	3	30	5/17	8	8	144	18/145
3	2	28	14/31	5	4	48	4/17	9	8	160	20/163
3	2	40	20/43	5	4	84	7/29	9	9	180	20/181
3	2	44	22/47	5	5	60	12/61	13	12	84	7/89
3	2	56	28/59	6	2	119	27/124	13	12	168	14/173
3	3	24	8/25	6	6	84	14/85	13	12	252	21/257
3	3	60	20/61	7	4	56	14/61	19	18	180	2/37
4	1	55	19/59	7	6	24	4/29	19	18	360	4/73
4	2	65	19/68	7	6	48	8/53	25	24	312	13/317
4	2	70	20/73	7	6	72	12/77	31	30	480	16/487
4	3	45	15/47	7	6	96	16/101	32	30	330	11/337
4	3	75	25/77	7	6	120	4/25	37	36	684	19/689

Table 3. The 42 triples of $\mathcal{O}$ covered by neither Theorem 7.1 ($q \leq 2m+1$) nor Theorem 7.3, with the exact value of ML$(V(m,r,w))$.

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